MILITARY SPECIFICATION

BLOWERS, SOOT (SHIPBOARD USE)

This amendment forms a part of Military Specification MIL-B-1947A(SHIPS), 18 August 1960.

Page 1, paragraph 1.2, listing, Class 1, line 2: Delete "600" and substitute "710".

Page 1, paragraph 2.1; Delete reference to all Federal Specifications.

Page 1, paragraph 2.1: Delete reference to Military Specifications MIL-B-233, MIL-B-10377, MIL-L-10547, MIL-P-15424 and MIL-C-16173 and substitute:

"MIL-V-22094 - Values, Globe, Angle and Y, Flanged Bonnet, Manually Operated (Sizes 2 Inches and Below)."

Page 1, paragraph 2.1: Delete reference to Military Standards , MIL-STD-129 and MIL-STD-167.

Page 2, paragraph 3.1: Delete and substitute:

"3.1 Head design characteristics. - For the head materials specified for each class of soot blowers (see 3.5.3.6.1 and 3.5.3.6.2), fluctuations of not more than 25°F. are permissible."

Page 2, paragraph 3.2: First sentence, delete and substitute "Class 1 soot blowers shall be furnished with ASA standard mounting flanges conforming to ASA Publication B16.5 and ASME Boiler and Pressure Vessel Code, Section VIII, for the maximum pressure and temperature specified"; line 8, delete "excess" and substitute "access".

Page 2: Add as paragraph 3.2.1.

"3.2.1 Flange finish. - The gasket contact surfaces of all flanges used in the soot-blower assembly (head, companion, orifice, element flange, and so forth) shall be finished with a circular lay (concentric or phonographic) having a roughness not exceeding 500 roughness height rating (RHR) produced by machining cuts of 0.003 inch maximum depth but not less than 40 cuts per inch of face width. The finish within the above limits selected by the manufacturer is to be designated on all applicable drawings."

Page 3, paragraph 3.5.2: Delete and substitute:

"3.5.2 Type I (stationary) soot blowers (operating mechanism). - Type I (stationary) soot blowers shall include an adjustable orifice that is preset by the supplier for 300 p.s.i. (or less as approved by the command or agency concerned) steam admission to the element. This orifice installation, or other suitable means to be installed in the vicinity, shall contain a 3/8 inch NPT tap for insertion of pressure measuring device to verify admission steam pressure entering element. Unless otherwise specified (see 6.1), type I soot blowers shall be complete with steam valves. Steam valves shall conform to MIL-V-22094, material to be suitable for the source of steam. A spring-loaded air relief valve of corrosion-resistant material (3/8 inch i.p.s. minimum) shall be provided to admit purging air to the element and to prevent combustion gases from

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entering the piping. A 3/8 inch connection shall be provided for this purpose. The air relief valve shall be provided with fittings suitable for connection to the pressurized air space by 1/2 inch steel or copper tubing. Unless otherwise specified (see 6.1), type I soot blowers shall be complete with air relief tubing."

Page 3, paragraph 3.5.3.1: Delete and substitute:

"3.5.3.1 Operation. - The steam head shall be designed to transmit steam to the element or lance from a fixed steam connection without steam leakage when the element or lance is being operated. It shall be designed so that the head inlet is on or below the horizontal in a plane normal to the axis of the soot blower element."

Page 3, paragraph 3.5.3.2, line 3: Delete "of the head."

Page 3, paragraph 3.5.3.3: Delete and substitute:

"3.5.3.3 Steam pressure control in head. - The method of steam admission and steam pressure control shall be as approved by the command or agency concerned. Means of pressure control shall be included in each head design to reduce the pressure at the element a maximum blowing pressure of 300 p.s.i. A 3/8 inch NPT tap is to be provided in each soot blower head for insertion of a pressure measuring device. This pressure control shall be an adjustable part of the head in order to effect no greater steam entrance pressure to the element than the above designated 300 p.s.i. The control shall be capable of external adjustment and shall be present by the supplier for the designated maximum."

Page 4, paragraph 3.5.3.5, last sentence: Delete and substitute "Unless otherwise specified (see 6.1), the scavenging air relief valves shall be connected to the pressurized air space by 1/2 inch steel or copper tubing and by suitable joints approved by the command or agency concerned."

Page 4, paragraph 3.5.3.6.1: Heading, delete "600" and substitute "710"; listing, delete reference to Head and substitute:

Page 4, paragraph 3.5.3.6.1 listing: Delete reference to Head and substitute:

"Head----------Carbon molybdenum alloy steel castings or forgings in accordance with ASME Publication SA-217 grade WC-1 or SA-182 grade F1, respectively."

Page 4, paragraph 3.5.3.6.2 listing: Delete reference to Head and substitute:

"Head----------1-1/4 percent chrome; 1/2 percent molybdenum alloy steel casting or forging in accordance with ASME Publication SA-217 grade WC-6 or SA-182 grade F11 respectively."

Page 4, paragraphs 3.5.3.7 and 3.5.3.8: Delete.

Page 5, paragraphs 3.5.4.2: Delete and substitute:

"3.5.4.2 Reduction gearing. - Reduction gearing shall be provided in the head for notation of the elements. The gear ratio shall be not less than 7 to 1. For manually operated units designed for rotation in one direction only, an arrow shall be cast on or attached to the gear wheel, the sheave wheel or operating handwheel, whichever is most convenient for observation by the operating personnel, to indicate the correct direction of rotation for operation of the blower. For power operated units, the correct direction of rotation for the element shall be indicated on the unit wherever it is most convenient."

Page 5: Add as paragraph 3.5.4.3:

"3.5.4.3 Connecting piece (swivel or extension tube). - A tubular connecting piece shall be provided between the head and the element. Corrosion-resisting material shall be used where the piece passes through the stuffing box in the head. The connecting piece shall be for the purpose of securing the element to the head during field installation in a suitable manner to insure maintenance of the proper angular position of the nozzles with respect to the operating mechanism in the head. An indicator to show the angular position of the nozzles to the operator at all times shall be attached to the gear wheel or secured by other means approved by the command or agency concerned. The swivel tube or the element shall be at least 6 inches longer than designed length to allow for field modification to suit installation conditions."

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Page 5: Add as paragraph 3.5.4.4:

"3.5.4.4 Head interchangeability. - All heads of like design within the same class shall have fully interchangeable internal parts. All replacement heads of types II and III shall be furnished with sufficient cam and trigger parts to allow their assembly to fit any rotation requirements, both 360 degrees and below. Replacement swivel tubes or elements supplied shall provide sufficient length to allow field modification to actual boiler depth conditions (at least 6 inches). Further, for all replacement heads means shall be provided to allow attachment of either 1-1/2 inch or 2-inch elements."

Page 5, paragraph 3.5.6, item (b): Delete and substitute:

"(b) Traveling carriage. - The traveling carriage mechanism shall be air-motor driven. Mechanical means shall be used to translate and rotate the lance. Manual emergency means shall be provided to allow the removal of the lance from boiler cavities in case of air-motor failure. For manual emergency retraction of lance, in addition to gearing with handwheel, suitable kits shall be furnished as necessary to allow rotation of the gearing into accessible position for continuation by shipbuilder from this point by rods and universal joints or by sheave wheel and chain for operation of emergency retract from front of boiler. The traveling carriage must be of rugged construction, with suitable means of supporting the lance, during operation of rotating and translating, to prevent whipping. Minimum rotation speed of the lance shall be five revolutions per minute. Minimum translation speed shall be 5 feet per minute. Suitable means shall be provided to mechanically restrain the feed tube from rotation in the steam chest. In addition, the feed tube shall be suitably ground and polished to reduce wear on the packing. A suitable means shall be provided to prevent, under all conditions of boiler or soot blower operation, excessive gas leakage around the lance where it passes through the boiler casing."

Page 5, paragraph 3.6.3, line 6: Delete "of" and substitute "by."

Page 7, paragraph 3.9: Delete and substitute:

"3.9 Drawings. - Drawings shall be furnished in accordance with MIL-D-963. An arrangement drawing showing the location of the soot blowers on the boiler and of the number and location of the nozzles in each element with respect to the tubes, as finally approved by the command or agency concerned, shall be furnished. Master sets, consisting of not more than six drawings per set, shall be prepared for such class and type of soot blower for which a supplier desires approval. These masters shall include all possible variations in construction. Commercial type accessories such as air-motors and accompanying gear shall be shown on one dimensioned assembly and major detail drawing, together with a list of material identifying by part and manufacturer the major replacement parts. Radiographic and magnetic particle inspection markings shall also be delineated on one of the drawings. On any contract or order, the previously approved master drawings indicated above shall serve as the nucleus for drawing preparation. When only types I and II soot blowers are required on a contract or order, total number of applicable drawings shall not exceed 6; number of drawings to cover all types of soot blowers on a contract or order shall not exceed 12."

Page 7, paragraph 3.11.1, item (h): Delete and substitute:

"(h) Element for hottest location in boiler."

Page 7, paragraph 3.11.1, item (i): Delete.

Page 9, paragraphs 4.6.4 and 4.6.5: Delete.

Page 9, paragraph 4.6.6.2: Delete and substitute:

"4.6.6.2 Magnetic particle inspection. - All pressure containing alloy forgings rated at 600 pounds and above shall be 100 percent magnetic particle inspected in accordance with MIL-STD-271. (Carbon steel pressure containing forgings are not to be magnetic particle inspected.)"

Pages 9 and 10, section 5: Delete and substitute:

"5. PREPARATION FOR DELIVERY

"5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. –

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"5.1.1 Soot blower components. -

"5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

"5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

"5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

"5.1.2 Onboard repair parts. - Onboard repair parts shall be unit protected in accordance with MIL-P-116. Parts shall be individually packaged unless used in sets or quantities greater than one.

"5.2 The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

("5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

"5.2.1.1 Preservation and packaging. -

"5.2.1.1.1 Level A. -

"5.2.1.1.1.1 Soot blower components. - Soot blower components (see 3.4 except 3.4(b) and (c)) shall be preserved and packaged method I in accordance with the guide lines of MIL-P-116. All external unpainted ferrous metal surfaces and all internal surfaces in contact with steam shall be coated with preservative compound conforming to grade 4 of MIL-C-16173. Antifriction bearing shall be filled to operating capacity with normal operating lubricant. All openings shall be plugged, capped or sealed to prevent entrance of dirt and moisture.

"5.2.1.1.1.2 Bearings (or hangers) and elements. - The bearings (or hangers) and elements shall be preserved and packaged method III in accordance with MIL-P-116.

"5.2.1.1.2 Level C. - Soot blower components, (see 3.4) except repair parts, shall be preserved and packaged in accordance with the supplier's commercial practice.

"5.2.1.2 Packing. -

"5.2.1.2.1 Level A. - Soot blower components, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Overseas
PPP-B-621	Class 2
PPP-B-640	Class 2

Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Shippping containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of the shipping containers shall not exceed 200 pounds. Shipping containers exceeding 200 pounds shall be modified by the addition of skids in accordance with the applicable box specification.

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"5.2.1.2.2 Level B. - Soot blower components, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Domestic
PPP-B-621	Class 1
PPP-B-640	Class 1

Caseliners will not be required. Closure shall be in accordance with the applicable box specification or appendix thereto.

"5.2.1.3 Marking. - In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

"5.2.2 Repair parts (onboard and stock). -

"5.2.2.1 Level A. - Repair parts, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 3
PPP-B-591	Class II
PPP-B-601	Overseas
PPP-B-621	Class 2
PPP-B-636	Weather-resistant
PPP-B-576	Class 2

Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 1-1/2 inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

"5.2.2.2 Level B. - Repair parts shall be packed in any of the boxes specified in 5.2.2.1 and as follows:

Specification	Type or class
PPP-B-585	Class 1 or 2, domestic
PPP-B-591	Class I
PPP-B-601	Domestic
PPP-B-621	Class 1
PPP-B-636	Weather-resistant
PPP-B-576	Class 1

Caseliners shall not be required. Closure of boxes shall be in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

"5.2.2.3 Repair parts boxes. - When specifically required in the contract or order, the on-board repair parts shall be packed in type M (metal) or type W (wood) boxes conforming to MIL-B-233 as specified (see 6.1). Repair parts boxes will not require overpacking for domestic shipment or storage but shall be overpacked in containers specified in 5.2.2.1 for overseas shipment.

"5.2.2.4 Index list of repair parts. - An index list of repair parts shall be inserted in each shipping container or repair parts box containing repair parts. The list shall contain a complete itemized listing of the box contents including stock numbers, nomenclatures and quantities. The list shall be placed in a waterproof bag or be so treated as to be resistant to water, oil and fading. The list shall be inserted in the index list support located on the interior side of the repair parts box cover, or suitably placed on the inside of the box for quick accessibility.

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"5.2.3 Anchoring, blocking, bracing and cushioning. - Anchoring, blocking, bracing and cushioning of the soot blower components and repair parts, when necessary, shall be provided to prevent damage during handling, shipment and storage.

"5.3 Tools. - Tools shall be preserved, packaged, and packed for the level specified (see 6.1) in accordance with MIL-P-15424.

"5.5 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)"

Page 10, paragraph 6.1, item (h): Delete and substitute:

"(h) Whether steam valves are required (see 3.5.2).
"(i) Whether tubing for air relief valves is required (see 3.5.2 and 3.5.3.5)."

Preparing activity:
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